

A CatBoost-Based Ablation Study of Scheduling Heuristics in Automobile Repossession

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Abstract—Automobile repossession is an operationally important component of the U.S. auto lending ecosystem, where recovery agencies must prioritize assignments under time and resource constraints. Industry guidance in the United States suggests that Sunday–Thursday nights represent the optimal window for automobile repossessions; however, this heuristic has not been empirically validated. Using the Automobile Repossession Dataset (ARD) comprising 203,436 attempts, we evaluate the incremental predictive value of weekday scheduling within a CatBoost framework under repeated stratified cross-validation. To contextualize the value of weekday information, we compare it against address-type features, which represent a more directly actionable assignment characteristic related to likely vehicle location and recovery conditions. Weekday, evaluated as a standalone temporal feature, produced no significant improvement in the primary metric Area Under the Precision-Recall Curve (AUPRC) under paired testing ($t(49) = 0.58$, $p = 0.5639$). Although small statistically significant increases were observed for Area Under the Receiver Operating Characteristic Curve (AUC-ROC) and Lift@10%, their magnitudes were negligible. In contrast, address-type features yielded substantially larger improvements in ranking performance. These results suggest that weekday scheduling practices have limited practical predictive value relative to structurally grounded assignment characteristics.

Index Terms—Auto repossession analytics, CatBoost, operational heuristics, lift analysis, SHAP interpretability

I. INTRODUCTION

When a borrower defaults on an automobile loan, the secured creditor retains the legal right to recover the collateral vehicle [1], generally by contracting an automobile repossession agency to retrieve it. Auto repossession is thus a critical enforcement mechanism within consumer auto lending markets. It is both high-volume—with >1.7 million repossessions annually in the U.S. [2]—and high-complexity, as information on vehicle whereabouts can be unreliable. Because assignments often contain several possible locations and must be prioritized under time and resource constraints, agencies frequently rely on operational heuristics, including assumptions about which days of the week offer the highest probability of successful recovery.

Repossession activities are commonly conducted overnight, when reduced traffic and pedestrian presence may lower operational risk. Within this broader nighttime strategy, practitioner commentary has suggested that recovery success may vary by day of the week. For example, CURepossession [3] describes

Sunday–Thursday nights as a potential “sweet spot” for recoveries, citing borrower routines, weekday work schedules, and increased vehicle availability at residential locations during early-week evenings. According to this perspective, vehicles may be more predictably located at borrower residences before weekend travel patterns or discretionary movement increases. Despite the operational intuition underlying this claim, it has not been formally evaluated using large-scale empirical data.

This study evaluates the incremental predictive contribution of weekday scheduling using the Automobile Repossession Dataset (ARD), comprising 203,436 assignments. We compare its contribution against address type, a more directly actionable assignment characteristic that reflects recovery context and likely vehicle location. Using a controlled CatBoost [4] modeling framework with repeated stratified cross-validation, we quantify modeling performance across feature configurations. To our knowledge, this study represents the first empirical evaluation of weekday-based operational heuristics in automobile repossession.

The remainder of this paper is structured as follows. Section II reviews related methodological work in imbalanced classification, gradient-boosted decision trees, and interpretable machine learning. Section III describes the ARD dataset and feature composition. Section IV outlines the experimental methodology, including preprocessing, cross-validation, and statistical testing procedures. Section V presents empirical results and SHapley Additive exPlanations (SHAP)-based interpretability analysis. Section VI concludes with operational implications and directions for future research.

II. RELATED WORK

Direct prior work at the intersection of automobile repossession and machine learning remains extremely limited. Consistent with our targeted search of publicly available literature from 2015–2025, we did not identify prior empirical studies applying machine learning to operational automobile repossession outcome prediction or weekday-based repossession heuristics. Accordingly, the most relevant prior works to the current study come from adjacent literatures on imbalanced classification, gradient-boosted decision trees, interpretable machine learning, and temporally informed decision-making.

A. Imbalanced Classification and Evaluation Metrics

The ARD exhibits moderate class imbalance (26.4% positive class). Prior research demonstrates that traditional threshold-dependent metrics such as classification accuracy may be misleading under imbalanced distributions [5], [6]. These studies further show that commonly reported measures such as Area Under the Receiver Operating Characteristic Curve (AUC-ROC) may be worse measures of true model performance than precision-recall-based metrics in severely imbalanced settings [7]. Precision-recall analysis has been shown to provide a more informative evaluation under imbalance, particularly when the positive class represents the operational event of interest. Area Under the Precision-Recall Curve (AUPRC) captures ranking performance across classification thresholds and is more sensitive to improvements in minority-class detection than Receiver Operating Characteristic (ROC)-based metrics [7].

Work by Johnson & Khoshgoftaar has examined evaluation strategies for imbalanced and large-scale tabular classification tasks [8]. These studies emphasize robust cross-validation procedures, metric selection aligned with operational objectives, and careful statistical testing when comparing model variants. These principles motivate our use of repeated stratified cross-validation [9], paired hypothesis testing, and AUPRC as the primary performance metric.

B. Gradient Boosted Decision Trees in Tabular Domains

Gradient-Boosted Decision Trees (GBDTs) [10] have emerged as state-of-the-art methods for structured tabular data. XGBoost [11] and CatBoost [4] have demonstrated strong predictive performance across financial, healthcare, and fraud detection domains [12].

CatBoost is particularly well-suited for mixed-type datasets due to its native handling of categorical variables via ordered target statistics, reducing prediction shift and overfitting in high-cardinality settings [4]. XGBoost provides flexible regularization and efficient parallelization, making it robust for large-scale classification tasks [11].

Recent comparative analyses indicate that GBDT models consistently outperform linear models on nonlinear tabular tasks, though gains may be modest when signal structure is primarily additive or low-dimensional [12]. Given the ARD’s mixture of categorical, binary, geospatial, and temporal features, GBDTs provide an appropriate modeling framework for isolating incremental predictive contributions of weekday and address-type variables.

C. Interpretability in Operational Machine Learning

Interpretability is particularly important in operational and risk-sensitive domains, where model decisions may affect resource allocation and therefore must remain transparent enough to support auditing, fairness review, and regulatory justification. SHapley Additive exPlanations (SHAP) provides a unified framework for estimating feature contributions in tree-based models [13]. SHAP values allow global ranking

of feature importance as well as local instance-level explanations [14].

In applied domains such as fraud detection and healthcare analytics, SHAP has been used to evaluate whether specific feature groups contribute meaningful signal beyond baseline predictors [14], [15]. We adopt SHAP analysis to quantify the relative contribution of weekday indicators compared to address-type and operational features within the ARD.

To our knowledge, no prior work has formally evaluated weekday-based operational heuristics within automobile repossession. While temporal patterns are widely studied in adjacent fields like predictive policing [16] and credit default [17], their application to the specific operational constraints of automobile repossession remains an unexplored frontier in the literature.

III. DATASET

A. Dataset Overview

The Automobile Repossession Dataset (ARD) [18], [19] is a proprietary, anonymized tabular dataset obtained from a regional U.S. repossession agency. The dataset contains 203,436 structured records spanning June 2021 through December 2024, with each row representing a unique repossession assignment. Each record corresponds to a distinct recovery assignment placed by a client.

The binary target variable, *status*, indicates whether the vehicle was successfully recovered (1) or not recovered (0) during the operational lifecycle of the assignment. Of all assignments, 53,801 (26.4%) resulted in successful recovery, while 73.6% did not. This establishes a moderately imbalanced binary classification setting.

The dataset contains 16 features across five operational categories, as summarized in Table I. All features reflect information available prior to or at the time of assignment scheduling. No post-recovery outcomes, agent notes, or post-disposition variables were included in model training. All personally identifiable information was removed prior to analysis, and entity names were anonymized.

B. Geospatial and Address Features

Geographic information is captured through a five-digit *zip* field and a categorical *address_type* variable. ZIP codes are treated as categorical identifiers, as numeric magnitude does not convey ordinal meaning. To reduce cardinality while preserving broad regional structure, ZIP codes were standardized to five digits and truncated to their three-digit prefixes, a level of geographic aggregation with precedent in facility-location and logistics modeling [20].

The *address_type* feature reflects the operational classification of the location associated with the assignment. Common categories include borrower home, additional addresses derived through skip tracing, co-borrower addresses, employer locations, and license plate recognition (LPR)-derived addresses.

Exploratory tabulations by address type showed substantially greater variation in raw recovery rates than was observed

TABLE I
ARD FEATURE ENGINEERING AND STRUCTURAL SUMMARY

Category	Feature Name	Type	Cardinality
Target	status	Binary	2
Geospatial	address_type	Categorical	17
	zip	Categorical	4214
Entity	client_name	Categorical	110
	lienholder_name	Categorical	858
	department_id	Categorical	7
Temporal	order_year/month/day	Numeric	4/12/31
Operational	assessed_score	Numeric	8
	repo_type	Categorical	13
	lpr_responses_processed	Binary	2
	touched_within_30_days	Binary	2
	vehicle_newer_than_10_years	Binary	2
	direct_client_designation	Binary	2
	title_record_24_months	Binary	2
	skip_address_verification	Binary	2

across weekdays. High-support categories such as “Borrower-Home” (26.74% on 163,881 assignments), “Additional Address” (25.04% on 15,569 assignments), and “DMV Address” (25.99% on 1,389 assignments) clustered near the overall baseline, whereas “LPR Location” exhibited a markedly higher raw success rate (45.02% on 2,341 assignments). Other categories displayed much lower recovery rates, including “Borrower-Work” (18.08%), “Co-Signer-Home” (16.72%), “Pickup Address” (1.81%), and “Unknown” (1.10%). The “On Hook” category showed an extremely high recovery rate (99.65%), indicating that this pre-recovery operational category is associated with unusually favorable recovery conditions, even though it is assigned before the final outcome is known. These descriptive results suggested that address type captures materially different recovery contexts and may provide stronger operational signal than coarse weekday scheduling alone, while also motivating a multivariate analysis to distinguish robust predictive value from simple descriptive association.

The distribution of address types is highly skewed. Approximately 80.6% of assignments correspond to “Borrower-Home” addresses, 7.7% to “Additional Address”, 5.9% to “Unknown”, 1.7% to “On Hook”, and 1.2% to “LPR Address”, with remaining categories aggregated as Other. The On Hook category refers to assignments in which the vehicle had already been physically located and was in immediate recovery position at the time the address classification was recorded; although strongly associated with successful outcomes, this category was not in one-to-one correspondence with the target label. Preliminary descriptive analysis revealed substantial heterogeneity in recovery rates across address categories, with success probabilities as low as $\sim 1\%$ for intelligence-poor categories (e.g., Unknown), while baseline borrower residence attempts averaged approximately 27%. This variation motivated evaluation of address type as a structurally meaningful comparator to weekday scheduling.

Address type is operationally meaningful because it directly influences vehicle visibility, likelihood of physical access, contact risk, and recovery logistics. As such, it represents a structurally grounded operational predictor and serves as a

natural comparator to weekday scheduling effects.

C. Assignment and Entity Features

Two high-cardinality categorical variables, `client_name` and `lienholder_name`, identify the originating entity and secured creditor, respectively. The dataset includes 110 distinct client entities, with the largest accounting for approximately 20% of all assignments, suggesting potential client-level heterogeneity in portfolio composition, geographic coverage, or operational constraints.

The variable `department_id` encodes one of seven internal operational divisions and was excluded from modeling because it reflects organizational routing rather than assignment characteristics.

D. Operational Indicators

The feature `assessed_score` represents an internally generated pre-assignment risk heuristic available at intake. The underlying scoring logic used is not transparent, limiting interpretability. However, we retain it as an operationally available predictor regardless. The feature `repo_type` represents the nature of the repossession assignment. Approximately 96% of assignments are involuntary repossessions requiring active vehicle location. Voluntary and impound assignments comprise a small minority. Impound assignments are treated as voluntary because the vehicle location is already identified by a third party.

Several binary indicators capture procedural and informational signals available prior to recovery attempts:

- `lpr_responses_processed`: whether the vehicle was previously detected via internal license plate recognition systems;
- `touched_within_30_days`: whether the assignment has been previously worked within the last 30 days;
- `vehicle_newer_than_10_years`: whether the vehicle’s age is ten years or less;
- `title_record_24_months`: whether title activity occurred within the last 24 months;

- `skip_address_verification`: whether the address on the assignment matches that of a third-party database;
- `direct_client_designation`: whether the assignment is considered a direct or forwarder client.

These features reflect informational recency, prior operational effort, and data verification status, all of which may influence recovery probability independent of scheduling practices.

E. Temporal Features

A derived `weekday` feature was constructed during preprocessing from `order_year`, `order_month`, and `order_day`. To treat weekdays as nominal categories rather than ordered values, we represented this as a categorical feature rather than an integer-coded ordinal feature. In ablation testing, `weekday` was evaluated as a standalone temporal feature to assess whether coarse day-of-week information contributed predictive value beyond the baseline model. Assignment volume was concentrated on business days, with Monday–Friday accounting for approximately 92% of all assignments. Among business days, the distribution was relatively balanced (ranging from 15.9% to 20.3% per day), while Saturdays (4.7%) and Sundays (3.3%) exhibited substantially lower volume.

Examining success rates by weekday, we observed only marginal differences between the lowest (24.51% on Saturdays) and highest (27.17% on Sundays) rates; rates for business days showed an even tighter spread of 26.24–26.85%. The small success-rate variance on weekends could reflect real differences in operational conditions, but could also be explained by random variability caused by the lower data volume on these days. This further motivates a controlled multivariate analysis of `weekday` as a predictive feature.

Collectively, the ARD represents a structured, operationally grounded dataset reflecting real-world assignment conditions, with heterogeneous entity, geographic, procedural, and temporal signals available prior to recovery execution.

IV. METHODOLOGY

A. Experimental Framework

Hyperparameters were fixed across all configurations to ensure controlled comparison across feature inclusion conditions. The CatBoost classifier was configured with 750 boosting iterations, tree depth of 7, and the default learning rate. These values were selected based on prior ARD modeling experiments [19], [21] to provide stable performance while avoiding configuration-specific optimization. Holding hyperparameters constant ensures that observed performance differences are attributable to feature inclusion rather than model optimization variance, consistent with controlled ablation practices in comparative modeling studies [9].

For our experiments, we establish three feature groups: Address Type, Calendar-Day (`order_year`, `order_month`, and `order_day`), and Weekday. In total, we evaluate five models: the “baseline” model using none of these feature groups (relying on the other eleven ARD features), three

models each using one of these feature groups, and a “full” model using all three feature groups.

B. Preprocessing

Missing values in the six binary operational indicators were imputed using Multivariate Imputation by Chained Equations (MICE) with a Random Forest estimator [22]; this method was selected to preserve potential multivariate dependency structure among features, and because of its strong performance in preliminary experiments. ZIP codes were truncated to three-digit regional prefixes to evaluate broader geographic effects. Categorical variables were passed directly to CatBoost, leveraging its native handling of categorical features through ordered target statistics to mitigate prediction shift [4].

Class imbalance was addressed using Random Undersampling (RUS) applied exclusively to training folds. This reduces majority-class dominance during model training, preventing model bias. No resampling was applied to validation folds, preserving natural class distribution during evaluation.

C. Evaluation Metrics

We report AUC-ROC, AUPRC, and Lift@10%. AUPRC was designated as the primary metric because it provides a more informative assessment of model performance in imbalanced settings where the minority class is the primary operational objective [7]. Unlike AUC-ROC, AUPRC focuses on the precision–recall trade-off and is more sensitive to improvements in the detection of successful recoveries. To quantify incremental performance differences across feature configurations, we report the change in each evaluation metric relative to the baseline model.

Lift@10% represents the model’s ability to concentrate successful outcomes within the highest-probability assignments. It was computed as the ratio of the positive rate within the top decile of predicted probabilities to the overall positive rate in the evaluation set [23]. For example, if the overall recovery rate is 26.4% and the recovery rate among the top 10% highest-scored assignments is 52.8%, then $\text{Lift@10\%} = 0.528 / 0.264 = 2.00$. In an operational context, this metric quantifies the incremental efficiency gain of a model-driven scheduling strategy compared to random assignment.

Each configuration was evaluated using ten runs of stratified five-fold cross-validation, yielding 50 trained models per configuration. Random seeds were varied across repetitions so that performance estimates were not tied to a single partitioning of the data, thereby reducing sensitivity to split-specific bias [9].

D. Statistical Testing

To evaluate incremental predictive contribution, paired two-sided *t*-tests were conducted across matched cross-validation folds ($n = 50$ per configuration) [24]. For each fold, metric values from alternative configurations were paired with the corresponding baseline fold results. Paired testing across matched folds controls for shared training partitions and reduces variance in model comparison [25]. Statistical significance was assessed at $\alpha = 0.05$. Although AUPRC was

TABLE II
PERFORMANCE SUMMARY ACROSS MODEL CONFIGURATIONS (MEAN $\pm$ SD OVER 50 FOLDS)

Configuration	AUC-ROC	AUPRC	Lift@10%
Baseline	0.7240 $\pm$ 0.0033	0.5001 $\pm$ 0.0042	2.2569 $\pm$ 0.0265
Weekday	0.7243 $\pm$ 0.0030	0.5002 $\pm$ 0.0042	2.2634 $\pm$ 0.0288
Calendar-Day	0.7294 $\pm$ 0.0031	0.5052 $\pm$ 0.0045	2.2874 $\pm$ 0.0291
Address Type	0.7673 $\pm$ 0.0026	0.5800 $\pm$ 0.0042	2.6131 $\pm$ 0.0394
Full Model	0.7720 $\pm$ 0.0026	0.5843 $\pm$ 0.0042	2.6424 $\pm$ 0.0275

TABLE III
PAIRED t -TEST RESULTS FOR AUPRC ACROSS KEY MODEL COMPARISONS

Comparison	Δ AUPRC	t -statistic	p -value
Weekday Only vs. Baseline	0.00014	0.58	0.5639
Calendar Day Only vs. Baseline	0.00511	18.81	< 0.001
Address Type Only vs. Baseline	0.07991	178.29	< 0.001
Full Model vs. Baseline	0.08423	186.09	< 0.001
Full Model vs. Address Type Only	0.00432	20.16	< 0.001

treated as the primary metric, paired tests were also conducted for AUC-ROC and Lift@10% to assess whether the inclusion of `weekday` produced measurable changes in secondary performance measures relative to the baseline model.

E. Interpretability Analysis

Feature contributions were evaluated using SHAP [13], providing a grounded framework for quantifying feature contributions to individual predictions based on Shapley values from cooperative game theory. For each trained model, absolute SHAP values were computed for validation instances using TreeExplainer [14]. Absolute SHAP values were averaged across validation instances and aggregated across cross-validation folds to produce global feature-importance rankings.

V. RESULTS

A. Classification Performance

Table II summarizes model performance across the evaluated configurations. To quantify the relative contribution of each feature group, Table III reports the incremental AUPRC differences between configurations, with paired t -tests to show significance. All results are aggregated across 50 repeated cross-validation runs.

Tables II and III show a consistent pattern across the evaluated configurations. When `weekday` was introduced as the only additional feature group, we found essentially no change in AUPRC relative to baseline ($\Delta = +0.00014$). A paired two-sided t -test across matched cross-validation runs confirmed that this difference was not statistically significant ($t(49) = 0.58$, $p = 0.5639$). Small statistically significant increases were observed for AUC-ROC ($\Delta = 2.58 \times 10^{-4}$, $t(49) = 2.22$, $p = 0.0313$) and Lift@10% ($\Delta = 0.00652$); however, these effects were negligible in magnitude.

Including raw calendar-day variables produced a modest improvement relative to baseline (AUPRC = 0.50520; $\Delta = +0.00511$). Corresponding gains were also observed in

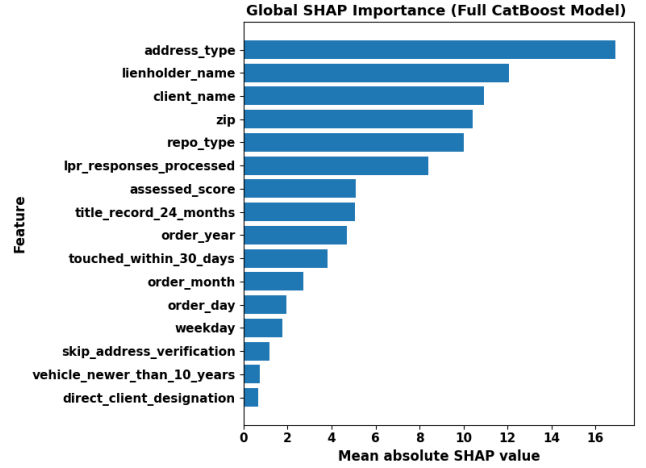


Fig. 1. Global feature importance (mean absolute SHAP values) for the CatBoost model using all features.

AUC-ROC ($\Delta = +0.00537$) and Lift@10% ($\Delta = +0.03045$), suggesting limited but measurable temporal signal at the day-of-month level. However, the magnitude of these effects remained small relative to address-based variables.

In contrast, inclusion of address-type information produced a substantial improvement. The address-only configuration increased AUPRC from 0.5001 to 0.5800 ($\Delta = +0.0799$), representing a 15.9% relative increase, with strong statistical significance ($p < 0.001$). Lift@10% improved from 2.2569 to 2.6131, a 15.8% relative increase. The magnitude of improvement from address-type inclusion was substantially larger than that observed for `weekday` inclusion.

The full model achieved the highest overall performance: AUC-ROC = 0.7720, AUPRC = 0.5843, and Lift@10% = 2.6424. Relative to baseline, this corresponds to a 17% relative increase in both AUPRC and Lift@10%. Importantly, the strongest performance was obtained when `weekday`, `calendar-day`, and `address-type` information were included jointly; however, the largest improvement was attributable to address-based features rather than `weekday` alone.

Collectively, these results indicate that `weekday` scheduling provides only negligible gains in both the primary precision-recall objective and secondary metrics, suggesting limited practical predictive value in repossession outcome modeling.

B. SHAP Analysis

Figure 1 presents the global feature-importance ranking. The most influential feature was `address_type`, followed by high-cardinality entity variables including `lienholder_name` and `client_name`. Additional meaningful contributors included `zip`, `repo_type`, `lpr_responses_processed`, and `assessed_score`, indicating that recovery predictions were driven primarily by assignment context, entity-level heterogeneity, and operational intelligence signals.

In contrast, the derived `weekday` feature ranked 13th out of 16 predictors by mean absolute SHAP value. This

indicates that weekday information contributed comparatively little to predicted recovery probability. Notably, the raw calendar variables (year, month, and day) each ranked above weekday, suggesting that finer-grained temporal information carries more predictive value than coarse day-of-week encoding alone. This confirms the findings in Section V-A that address information was a major source of predictive signal, while weekday information contributed only marginally.

VI. CONCLUSION

This study evaluated the predictive value of weekday-based scheduling cues for auto repossession success prediction, using the Automobile Repossession Dataset (ARD) within a controlled CatBoost ablation framework. While industry sources assert that Sunday–Thursday nights represent optimal recovery windows [3], paired comparisons found that adding weekday information to a baseline classification model does not improve AUPRC ($p = 0.5639$). Although statistically significant gains were observed for AUC-ROC and Lift@10%, the corresponding effect sizes were negligible. This finding holds despite existing institutional scheduling preferences reflected in the ARD, biasing data towards Sunday–Thursday nights overall.

In contrast, address-based features provided measurable incremental discrimination, with the inclusion of `address_type` increasing AUPRC by 0.0799 and AUC-ROC by 0.0433 relative to the baseline model. From an operational standpoint, these results suggest that resource prioritization strategies should emphasize structural assignment characteristics (e.g., address context and operational indicators) rather than relying on weekday-based scheduling assumptions.

Future work should investigate whether other temporal features provide greater predictive value than weekday alone. Examples include time-of-day effects, holiday timing, and interactions between temporal variables and address-based characteristics. Such analyses may help determine whether the limited contribution observed for weekday reflects the weakness of scheduling signal overall, or whether more targeted temporal formulations may provide greater operational value.

This study provides empirical evidence that weekday-based scheduling offers limited practical predictive value in multivariate models of repossession success. Address-based contextual features, however, materially improve discrimination and ranking performance. These findings support a transition from heuristic scheduling practices toward empirically validated, data-driven assignment prioritization frameworks.

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